

III. Visualization using Power BI

1. Data Loading and Initial Transformation

Step 1 : Open the Power BI app and click on 'Get Data' in Home Tab.

Step 2 : Choose Text/Csv if your data is in .csv format or excel if the dataset is in .xlsx format.

Step 3 : Navigate to your file location and select the dataset you wish to load.

Step 4 : Power BI will automatically read the dataset and display a preview of the data.

Step 5 : Select **Load** if no transformation is needed and choose **Transformation** for data cleaning like removing unnecessary columns , removing duplicates, filling null values and changing data types.

Step 6 : After doing necessary transformations click on 'Close and Apply'.

2. Data Transformation

Create Measures :

- a. To insert a new measure click on 'New Measure' in the Home Tab.
- b. Create measures like
 - Average Discover Weekly Engagement =
`AVERAGE(Global_Music_Streaming_Listener_Preferences[Discover Weekly Engagement (%)])`
 - Average Minutes Streamed per Day =
`AVERAGE(Global_Music_Streaming_Listener_Preferences[Minutes Streamed Per Day])`
 - Average Repeat Song Rate =
`AVERAGE(Global_Music_Streaming_Listener_Preferences[Repeat Song Rate (%)])`

- AvgMinutesByGenre = `CALCULATE(AVERAGE(Global_Music_Streaming_Listener_Preferences[Minutes Streamed Per Day]),ALLEXCEPT(Global_Music_Streaming_Listener_Preferences, Global_Music_Streaming_Listener_Preferences[Top Genre]))`
- Total User = `DISTINCTCOUNT(Global_Music_Streaming_Listener_Preferences[User_ID])`
- Countryflag = Switch(

'Global_Music_Streaming_Listener_Preferences'[Country],

"Australia", "https://flagcdn.com/w80/au.png",

"South Korea", "https://flagcdn.com/w80/kr.png",

"UK", "https://flagcdn.com/w80/gb.png",

"France", "https://flagcdn.com/w80/fr.png",

"Canada", "https://flagcdn.com/w80/ca.png",

"Germany", "https://flagcdn.com/w80/de.png",

"Japan", "https://flagcdn.com/w80/jp.png",

"India", "https://flagcdn.com/w80/in.png",

"Brazil", "https://flagcdn.com/w80/br.png",

"USA", "https://flagcdn.com/w80/us.png",

Blank()

)

Data Category: Set it as Image URL

